Tao CHENG

Shenzhen / Hangzhou, China | travischeng123@gmail.com | tao-cheng.cc

Education

Zhejiang University 2023.09 — 2027.06
B.Eng. in Computer Science and Technology (expected June 2027) *Hangzhou, China*

- **Academic Performance**: GPA 4.78/5.00 (93.46/100) | Rank 1/281 in cohort
- **Scholarships**: National Scholarship (2025, 2026) | NITORI International Scholarship (2024)

Research Interests

3D Vision, World Models, Embodied AI.

Publications

** denotes equal contribution; † denotes corresponding author.*

DiffWind: Physics-Informed Differentiable Modeling of Wind-Driven Object Dynamics
Y. Lei*, B. Zhao*, Z. Yang*, X. Li, **T. Cheng**, H. Peng, …, Z. Cui† *ICLR 2026*
DiffWind jointly reconstructs wind fields and deformable object dynamics from sparse-view videos through differentiable physics, enabling simulation under new wind conditions.

PhysSkin: Real-Time and Generalizable Physics-Based Animation via Self-Supervised Neural Skinning
Y. Lei, **T. Cheng**, X. Li, B. Zhao, S. Huang, R. Hu, …, Z. Cui† *CVPR 2026* ***Highlight***
PhysSkin learns neural skinning fields from static 3D geometry through self-supervision, enabling real-time physics-based animation across unseen shapes and spatial discretizations.

PhysSkin++: Real-Time and Generalizable Physics-Based Animation via Self-Supervised Neural Skinning
Y. Lei, B. Liu, X. Li, **T. Cheng**, Z. Wang, B. Zhao, …, Z. Cui† *arXiv preprint 2026 (under review at IEEE TPAMI)*
PhysSkin++ learns generalizable neural skinning fields from static geometry, enabling real-time physics-based animation with interactive control, collision-aware interactions, and rig-complementary deformation.

Project Experience

SRTP: City-scale Multi-modal 3D Reconstruction & Weather Simulation 2025.04 — 2026.05
Project Leader | National Level Research Grant *State Key Lab of CAD&CG, ZJU*

- **Developed** a framework integrating multimodal observations for consistent 3D scene reconstruction and rendering.
- **Contributed** to research projects on physics-based animation and wind-driven dynamics.

"Shenzhen Cup" National Collegiate Mathematical Modeling Challenge 2024.04 — 2024.08
Team Leader | Second Prize (3rd Place Nationally) *Shenzhen, China*

- **Developed** an optimization algorithm for sonic boom localization of rocket debris, enhancing positioning accuracy.
- **Led** a team of three to deliver high-quality technical reports, securing a 10,000 CNY prize for top-tier performance.

Campus Experience

HPC101: High-Performance Computing Workshop 2024.07 — 2024.08

- **Mastered** cluster configuration, parallel computing, and CUDA programming.
- **Optimized** CPUBench on Kunpeng CPUs (PAC 2024) via compiler tuning and C/Fortran profiling.

Data Factor Markets Workshop 2025.06 — 2025.07

- **Mastered** core concepts including Game Theory, MAB algorithms, and auction mechanism design for data assets.
- **Implemented** QIRANA's (SIGMOD '17) query-based data pricing algorithm, achieving full marks for the project.

Skills

- **Programming**: Python, C++, CUDA, LaTeX, Typst
- **Research Tools**: PyTorch, Git, Blender, Mamba, Codex
- **Languages**: English (IELTS 7.5), Chinese (native)